

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3052

3 Credits: 3

Program Core

Source-grounded

Manufacturing Technology for

□ To provide a basic knowledge on metal casting and fabrication processes.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3052 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3052.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	3052
Course title	Manufacturing Technology for
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

40 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a basic knowledge on metal casting and fabrication processes.
- To acquire knowledge of machine tools for specific applications, metal forming, additive manufacturing and their applications.

Course outcomes

CO	Official outcome	Study level
CO1	Choose metal casting and powder metallurgy. 12 Applying Classify sheet metal works used in automobile	See official syllabus
CO2	manufacturing and relate a fabrication process for 11 Understanding specific automobile application. Demonstrate the working of standard machine tools such as lathe, shaping, milling, drilling	See official syllabus
CO3	11 Understanding machines and the basics of flexible manufacturing system. Outline the application of forging, rolling and	See official syllabus
CO4	additive manufacturing techniques in automobile 9 Understanding industry.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Choose metal casting and powder metallurgy.	12	As specified
Module 2	particular fabrication process for specific automobile application.	11	As specified
Module 3	milling, drilling machine and the basics of flexible manufacturing system.	8	As specified
Module 4	:	9	As specified

MODULE 1

Choose metal casting and powder metallurgy.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Choose metal casting and powder metallurgy.
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

▼ Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material

Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recall casting and pattern. 1 remembering outline the requirements of a pattern material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റീകാസ്റ്റിംഗ് ആന്റ് പാറ്റേൺ. 1 Remembering Outline the requirements of a pattern material റീകാസ്റ്റിംഗ് പാറ്റേൺ definition/meaning റീകാസ്റ്റിംഗ് പാറ്റേൺ. റീകാസ്റ്റിംഗ് പാറ്റേൺ റീകാസ്റ്റിംഗ്, steps, application റീകാസ്റ്റിംഗ് പാറ്റേൺ റീകാസ്റ്റിംഗ്. Technical terms English-റീകാസ്റ്റിംഗ് പാറ്റേൺ റീകാസ്റ്റിംഗ്.

▼ 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece

1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and list different types of pattern materials. classify patterns like single, split, loose piece should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece definition/meaning. , steps, application . Technical terms English- .

▼ 1 Understanding and sweep pattern. Define pattern allowance and define pattern

1 Understanding and sweep pattern. Define pattern allowance and define pattern is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and sweep pattern. Define pattern allowance and define pattern using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and sweep pattern. define pattern allowance and define pattern should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding and sweep pattern. Define pattern allowance and define pattern means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding and sweep pattern. Define pattern allowance and define pattern definition/meaning. 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 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2608

▼ allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core

allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, allowances such as shrinkage, machining, 1 understanding draft, distortion, shake allowances. define mould, cope, drag, runner, riser, core should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core definition/meaning. steps, application. Technical terms English-
 definition/meaning. steps, application. Technical terms English-

▼ 1 Understanding and chaplets. Distinguish green sand moulding and dry sand

1 Understanding and chaplets. Distinguish green sand moulding and dry sand is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and chaplets. Distinguish green sand moulding and dry sand using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and chaplets. distinguish green sand moulding and dry sand should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding and chaplets. Distinguish green sand moulding and dry sand means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-1 Understanding and chaplets. Distinguish green sand moulding and dry sand പരിച്ഛേദനം പരിച്ഛേദനം definition/meaning പരിച്ഛേദനം. പരിച്ഛേദനം പരിച്ഛേദനം പരിച്ഛേദനം, steps, application പരിച്ഛേദനം പരിച്ഛേദനം പരിച്ഛേദനം. Technical terms English-1 പരിച്ഛേദനം പരിച്ഛേദനം.

▼ moulding and identify automotive components 1 Applying made by these processes. Explain hot chamber die casting, cold chamber die casting, centrifugal casting and

moulding and identify automotive components 1 Applying made by these processes. Explain hot chamber die casting, cold chamber die casting, centrifugal casting and is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify moulding and identify automotive components 1 Applying made by these processes. Explain hot chamber die casting, cold chamber die casting, centrifugal casting and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, moulding and identify automotive components 1 applying made by these processes. explain hot chamber die casting, cold chamber die casting, centrifugal casting and should be discussed with attention

to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What moulding and identify automotive components 1 Applying made by these processes. Explain hot chamber die casting, cold chamber die casting, centrifugal casting and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-മെറ്റൽ മൗൾഡിംഗ് ആൻഡ് ഓട്ടോമോട്ടീവ് കമ്പോൺറന്റുകൾ 1 ആപ്ലൈ ചെയ്തതാൽ ഉണ്ടാക്കിയതാൽ. Explain hot chamber die casting, cold chamber die casting, centrifugal casting and മെറ്റൽ മൗൾഡിംഗ്. മെറ്റൽ മൗൾഡിംഗ് മെറ്റൽ മൗൾഡിംഗ്, steps, application മെറ്റൽ മൗൾഡിംഗ് മെറ്റൽ മൗൾഡിംഗ്. Technical terms English-ൽ മെറ്റൽ മൗൾഡിംഗ് മെറ്റൽ മൗൾഡിംഗ്.

▼ 1 Applying identify automotive components made these processes.

1 Applying identify automotive components made these processes. is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying identify automotive components made these processes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying identify automotive components made these processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Applying identify automotive components made these processes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Applying identify automotive components made these processes. definition/meaning. , steps, application . Technical terms English- .

▼ Define various casting defects. 1 Understanding Define powder metallurgy and classify its

Define various casting defects. 1 Understanding Define powder metallurgy and classify its is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define various casting defects. 1 Understanding Define powder metallurgy and classify its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define various casting defects. 1 understanding define powder metallurgy and classify its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Define various casting defects. 1 Understanding Define powder metallurgy and classify its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Define various casting defects. 1 Understanding Define powder metallurgy and classify its definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 1 Understanding merits and demerits . Illustrate the methods for making of metal

1 Understanding merits and demerits . Illustrate the methods for making of metal is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding merits and demerits . Illustrate the methods for making of metal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding merits and demerits . illustrate the methods for making of metal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding merits and demerits . Illustrate the methods for making of metal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding merits and demerits . Illustrate the methods for making of metal definition/meaning . steps, application . Technical terms English- .

▼ powders such as atomization, reduction, 1 Understanding electrolysis and spotting. Outline various steps and secondary

powders such as atomization, reduction, 1 Understanding electrolysis and spotting. Outline various steps and secondary is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify powders such as atomization, reduction, 1 Understanding electrolysis and spotting. Outline various steps and secondary using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, powders such as atomization, reduction, 1 understanding electrolysis and spotting. outline various steps and secondary should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What powders such as atomization, reduction, 1 Understanding electrolysis and spotting. Outline various steps and secondary means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പുറം powders such as atomization, reduction, 1 Understanding electrolysis and spotting. Outline various steps and secondary പരമ്പരാഗതമായ രീതികൾ definition/meaning പരമ്പരാഗതമായ രീതികൾ. പരമ്പരാഗതമായ രീതികൾ, steps, application പരമ്പരാഗതമായ രീതികൾ. Technical terms English-ൽ പരമ്പരാഗതമായ രീതികൾ.

▼ 1 Understanding operations in powder metallurgy process. Identify automobile components made by Applying

1 Understanding operations in powder metallurgy process. Identify automobile components made by Applying is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding operations in powder metallurgy process. Identify automobile components made by Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding operations in powder metallurgy process. identify automobile components made by applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding operations in powder metallurgy process. Identify automobile components made by Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding operations in powder metallurgy process. Identify automobile components made by Applying definition/meaning. , steps, application . Technical terms English- .

▼ **1 powder metallurgy. Materials For Automobiles And Metal Casting**
Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a

1 powder metallurgy. Materials For Automobiles And Metal Casting
Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a is an official topic in Module 1 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 powder metallurgy. materials for automobiles and metal casting introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. moulding terminology such as cope, drag, runner, riser, core and chaplets. definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. discuss various casting defects. powder metallurgy steps, secondary operations. name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. classify sheet metal works used in automobile field and relate a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 powder metallurgy. Materials For Automobiles And Metal Casting Introduction to metal castings and patterns, pattern materials, types of patterns-single, split, loose piece, sweep, pattern allowances such as shrinkage, machining, draft, distortion, shake allowances. Moulding terminology such as cope, drag, runner, riser, core and chaplets. Definition green sand and dry sand moulding, permanent mould casting such as hot chamber and cold chamber die casting and centrifugal casting and identify automotive components made these processes. Discuss various casting defects. Powder metallurgy steps, secondary operations. Name the automotive components made by powder

metallurgy, merits and demerits of powder metallurgy. Classify sheet metal works used in Automobile field and relate a പൊതുവായ പദപരിചയം definition/meaning പൊതുവായ പദപരിചയം. പൊതുവായ പദപരിചയം, steps, application പൊതുവായ പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

particular fabrication process for specific automobile application.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: particular fabrication process for specific automobile application.
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and

1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding application of sheet metal in automobile. list metals used for sheet metal work and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ **1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking,**

1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding standard gauge numbers used for sheet metals. define various sheet metal operations like - shearing, cutting off, parting, blanking, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking, definition/meaning. steps, application. Technical terms English- .

▼ 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing.

1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding punching, notching, slitting, lancing, bending, drawing and squeezing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 2-1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing. പരമ്പരാഗതമായി ഉപയോഗിക്കുന്ന പദങ്ങൾ definition/meaning ഉൾപ്പെടെയുള്ളവ. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ. Technical terms English-ൽ പദങ്ങൾ പദങ്ങൾ.

▼ Define welding and weldability. 1 Remembering Explain the features of welding and general

Define welding and weldability. 1 Remembering Explain the features of welding and general is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define welding and weldability. 1 Remembering Explain the features of welding and general using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define welding and weldability. 1 remembering explain the features of welding and general should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Define welding and weldability. 1 Remembering Explain the features of welding and general means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Define welding and weldability. 1 Remembering Explain the features of welding and general definition/meaning. steps, application. Technical terms English-
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▼ 1 Understanding classification of welding. List the equipment used in arc welding and

1 Understanding classification of welding. List the equipment used in arc welding and is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding classification of welding. List the equipment used in arc welding and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding classification of welding. list the equipment used in arc welding and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding classification of welding. List the equipment used in arc welding and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding classification of welding. List the equipment used in arc welding and its definition/meaning. Explain the steps, application of arc welding. Technical terms English- Malayalam.

▼ 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and

1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding discuss the arc welding principle. outline mig welding, tig welding and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding discuss the arc welding principle. Outline MIG Welding, TIG Welding and CO_2 Arc Welding definition/meaning CO_2 Arc Welding. CO_2 Arc Welding CO_2 Arc Welding, steps, application CO_2 Arc Welding CO_2 Arc Welding. Technical terms English- CO_2 Arc Welding CO_2 Arc Welding.

▼ 1 Understanding Submerged arc welding. List equipment for gas welding and gas

1 Understanding Submerged arc welding. List equipment for gas welding and gas is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Submerged arc welding. List equipment for gas welding and gas using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding submerged arc welding. list equipment for gas welding and gas should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding Submerged arc welding. List equipment for gas welding and gas means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-1 Understanding Submerged arc welding. List equipment for gas welding and gas definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 1 Understanding cutting. Illustrate oxy-acetylene welding and types of

1 Understanding cutting. Illustrate oxy-acetylene welding and types of is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding cutting. Illustrate oxy-acetylene welding and types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding cutting. illustrate oxy-acetylene welding and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding cutting. Illustrate oxy-acetylene welding and types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding cutting. Illustrate oxy-acetylene welding and types of definition/meaning steps, application Technical terms English-

▼ 1 Understanding flames. Explain resistance welding such as - spot,

1 Understanding flames. Explain resistance welding such as - spot, is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding flames. Explain resistance welding such as - spot, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding flames. explain resistance welding such as - spot, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding flames. Explain resistance welding such as – spot, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding flames. Explain resistance welding such as – spot, definition/meaning. Principle, components, steps, application. Technical terms English- Malayalam.

▼ 1 Understanding butt, seam welding.

1 Understanding butt, seam welding. is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding butt, seam welding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding butt, seam welding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding butt, seam welding. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding butt, seam welding.
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ **Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping,**

Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, is an official topic in Module 2 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comparesoldering and brazing. 1 understanding sheet metal and fabrication processes introduction to sheet metal work and applications, metals used for sheet metal work, standard gauge numbers. sheet metal operations-shearing, cutting off, parting, blanking, punching, notching, slitting, lancing, bending, drawing and squeezing. introduction to welding - classification of welding such as fusion welding and solid state welding. - arc welding principle and equipment, types of arc welding - mig welding, tig welding, submerged arc welding. equipment for gas welding and gas cutting. oxy - acetylene welding and types of flames. principle of resistance welding - spot, butt, seam welding. comparison of soldering and brazing. demonstrate the working of standard machine tools like lathe, shaping, should be discussed with

attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Comparesoldering and brazing. 1 Understanding Sheet Metal And Fabrication Processes Introduction to Sheet Metal Work and Applications, metals used for Sheet Metal work, Standard Gauge numbers. Sheet metal operations-shearing, cutting off, Parting, blanking, Punching, notching, slitting, Lancing, Bending, Drawing and squeezing. Introduction to Welding - Classification of welding such as fusion

welding and solid state welding. - Arc welding principle and equipment, Types of Arc welding - MIG Welding, TIG Welding, Submerged arc welding. Equipment for Gas welding and gas cutting. Oxy - acetylene welding and Types of flames. Principle of resistance welding - Spot, butt, seam Welding. Comparison of soldering and Brazing. Demonstrate the working of standard machine tools like lathe, shaping, ഏകദേശരേഖാ വരവ് definition/meaning ഏകദേശരേഖാ വരവ്. ഏകദേശരേഖാ വരവ് steps, application ഏകദേശരേഖാ വരവ്. Technical terms English-ഏകദേശരേഖാ വരവ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

milling, drilling machine and the basics of flexible manufacturing system.

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: milling, drilling machine and the basics of flexible manufacturing system.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ **their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,**

their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, their functions, working and lathe 2 understanding specifications. define various lathe operations such as facing, centering, drilling, plain turning, boring, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,
 definition/meaning, steps, application Technical terms English-
 .

▼ reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism.

reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, reaming, chamfering taper turning and thread cutting. identify the components of a of shaper and demonstrate the crank and slotted lever type 1 understanding m3.03 quick return motion mechanism. should be

discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism. definition/meaning .
 , steps, application .
 Technical terms English- .

▼ Identify the components of a drilling machine.

Identify the components of a drilling machine. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the components of a drilling machine. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the components of a drilling machine. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the components of a drilling machine. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓർമ്മപ്പെടുത്തുക Identify the components of a drilling machine. ഓർമ്മപ്പെടുത്തുക definition/meaning ഓർമ്മപ്പെടുത്തുക. ഓർമ്മപ്പെടുത്തുക ഓർമ്മപ്പെടുത്തുക, steps, application ഓർമ്മപ്പെടുത്തുക ഓർമ്മപ്പെടുത്തുക. Technical terms English-ഓർമ്മപ്പെടുത്തുക ഓർമ്മപ്പെടുത്തുക.

▼ Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and

Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define drilling, reaming and boring. 1 understanding identify the components of a column and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and definition/meaning. steps, application. Technical terms English- .

▼ 2 Understanding knee type milling machine. Define plain milling, side milling, face

2 Understanding knee type milling machine. Define plain milling, side milling, face is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding knee type milling machine. Define plain milling, side milling, face using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding knee type milling machine. define plain milling, side milling, face should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding knee type milling machine. Define plain milling, side milling, face means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding knee type milling machine. Define plain milling, side milling, face definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 1 Understanding milling, angle milling, end milling.

1 Understanding milling, angle milling, end milling. is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding milling, angle milling, end milling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding milling, angle milling, end milling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding milling, angle milling, end milling. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 1 Understanding milling, angle milling, end milling. ഫലനം definition/meaning. ഫലനം ഫലനം, steps, application ഫലനം ഫലനം. Technical terms English- ഫലനം ഫലനം.

▼ Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system

Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the components of a CNC machine. 1 Understanding Outline the Flexible manufacturing system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the components of a cnc machine. 1 understanding outline the flexible manufacturing system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

▼ **1 Understanding and relate its applications. Machine Tools And Operations** Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing

1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing is an official topic in Module 3 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding and relate its applications. machine tools and operations lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. hole making operations - drilling - reaming - boring. milling - column and knee type milling machine - various operations performed on milling machine. constructional features of cnc machines, illustrate flexible manufacturing system and its applications. outline the application of forging, rolling and additive manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 1 Understanding and relate its applications. Machine Tools And Operations Lathe - constructional features and working, various lathe operations such as facing, centering, drilling, plain turning, boring, reaming, chamfering taper turning and thread cutting. - Shaper - constructional features and working of the crank and slotted lever type quick return motion mechanism - constructional features of a drilling machine. Hole making operations - drilling - reaming - boring. Milling - Column and knee type milling machine - various operations performed on milling machine. Constructional features of CNC machines, illustrate flexible manufacturing system and its applications. Outline the application of forging, rolling and additive manufacturing

definition/meaning ഉപയോഗിക്കുന്നതിനുള്ള. ഉപയോഗിക്കുന്നതിനുള്ള, steps, application ഉപയോഗിക്കുന്നതിനുള്ള ഉപയോഗിക്കുന്നതിനുള്ള. Technical terms English-ഉപയോഗിക്കുന്നതിനുള്ള ഉപയോഗിക്കുന്നതിനുള്ള.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

This module is presented from the official syllabus scope for Manufacturing Technology for. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ general classification of metal forming 1 Understanding operations.

general classification of metal forming 1 Understanding operations. is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify general classification of metal forming 1 Understanding operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, general classification of metal forming 1 understanding operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What general classification of metal forming 1 Understanding operations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന general classification of metal forming 1 Understanding operations. എന്നൊക്കെ definition/meaning എന്നൊക്കെ. എന്നൊക്കെ എന്നൊക്കെ, steps, application എന്നൊക്കെ എന്നൊക്കെ എന്നൊക്കെ. Technical terms English-എന്ന എന്നൊക്കെ എന്നൊക്കെ.

▼ Define forging. 1 Understanding Define forging, smith forging and machine

Define forging. 1 Understanding Define forging, smith forging and machine is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define forging. 1 Understanding Define forging, smith forging and machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define forging. 1 understanding define forging, smith forging and machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Define forging. 1 Understanding Define forging, smith forging and machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Define forging. 1 Understanding Define forging, smith forging and machine definition/meaning. steps, application. Technical terms English-.

▼ 1 Understanding forging Demonstrate the machine forging of crank

1 Understanding forging Demonstrate the machine forging of crank is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding forging Demonstrate the machine forging of crank using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding forging demonstrate the machine forging of crank should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding forging Demonstrate the machine forging of crank means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Understanding forging Demonstrate the machine forging of crank definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill

1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding shaft and connecting rod. illustrate rolling and rolling mill should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill definition/meaning. principle, components or stages, steps, application. Technical terms English- Malayalam.

▼ 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous

1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding configurations. define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Understanding configurations. Define direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion

1 Understanding extrusion, discrete extrusion and hydrostatic extrusion is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding extrusion, discrete extrusion and hydrostatic extrusion should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Understanding extrusion, discrete extrusion and hydrostatic extrusion എന്നിങ്ങനെ definition/meaning ഉൾപ്പെടെ. എന്നാൽ ഏതു ഏതു, steps, application ഉൾപ്പെടെ ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

▼ Define wire drawing and tube drawing.

Define wire drawing and tube drawing. is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define wire drawing and tube drawing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define wire drawing and tube drawing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define wire drawing and tube drawing. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Define wire drawing and tube drawing.
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English- നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications,

Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define hydroforming and shot peening. 1 understanding summarize the fundamentals, applications, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Define hydroforming and shot peening. 1 Understanding Summarize the fundamentals, applications, definition/meaning. definition/meaning. steps, application. Technical terms English- Malayalam.

▼ **classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalkpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 <http://www.youtube.com/watch?v=TeBX6cKKHWY>**

classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd

Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 <http://www.youtube.com/watch?v=TeBX6cKKHWY> is an official topic in Module 4 of Manufacturing Technology for. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2

www.youtube.com/watch?v=T5gjkYvMg8A 3

<http://www.youtube.com/watch?v=TeBX6cKKHWY> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classifications of additive manufacturing and 1 understanding define rapid prototyping. contents metal forming and powder metallurgy hot working, cold working and classification of metal forming operations. forging- smith forging and machine forging. machine forging of crank shaft and connecting rod. rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. wire drawing, tube drawing, hydroforming, shot peening, fundamentals, classifications and applications of additive manufacturing and rapid prototyping. text / reference: t/r book title/author elements of workshop technology vol-i manufacturing process edition-by t1 hajra choudry. elements of workshop technology vol-ii manufacturing process edition-by t2 hajra choudry r1 work shop technology by r.s khurmi & j.k gupta of s.chand &co. ltd manufacturing engineering and technology by seropekalpakjian, steven r r2 schmid pearson education-delhi r3 manufacturing technology-1by p.c sharma of s.chand publications. r4 engineering materials by er. r.k.rajpoot of s.chand publications online resources: sl.no website link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=t5gjkYvMg8A 3 <http://www.youtube.com/watch?v=tebx6ckkhwy> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder

Aspect	What to prepare
	Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalkpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 http://www.youtube.com/watch?v=TeBX6cKKHWY means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ classifications of additive manufacturing and 1 Understanding define rapid prototyping. Contents Metal Forming And Powder Metallurgy Hot working, cold working and classification of metal forming operations. Forging- smith forging and machine forging. Machine forging of crank shaft and connecting rod. Rolling and rolling mill

configurations. direct extrusion, indirect extrusion, hot extrusion, cold extrusion, continuous extrusion, discrete extrusion, hydrostatic extrusion. Wire drawing, tube drawing, hydroforming, shot peening, Fundamentals, classifications and applications of additive manufacturing and rapid Prototyping. Text / Reference: T/R Book Title/Author Elements of Workshop Technology Vol-I Manufacturing Process edition-By T1 Hajra Choudry. Elements of Workshop Technology Vol-II Manufacturing Process edition-By T2 Hajra Choudry R1 Work shop technology By R.S KHURMI & J.K GUPTA of S.CHAND &Co. Ltd Manufacturing Engineering and Technology by Seropekalkpakjian, Steven R R2 Schmid Pearson Education-Delhi R3 Manufacturing Technology-1By P.C Sharma of S.CHAND Publications. R4 Engineering Materials by Er. R.K.RAJPUT of S.CHAND Publications Online Resources: Sl.No Website Link 1 www.nptel.ac.in/courses/112105126/36 2 www.youtube.com/watch?v=T5gjkYvMg8A 3 <http://www.youtube.com/watch?v=TeBX6cKKHWY> ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഏകദേശം ഏകദേശം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material. (3 marks)

► Define or explain 1 Understanding and list different types of pattern materials. Classify patterns like single, split, loose piece. (3 marks)

► Define or explain 1 Understanding and sweep pattern. Define pattern allowance and define pattern. (3 marks)

► Define or explain allowances such as shrinkage, machining, 1 Understanding draft, distortion, shake allowances. Define mould, cope, drag, runner, riser, core. (3 marks)

Module 2

► Define or explain 1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and. (3 marks)

► Define or explain 1 Understanding standard gauge numbers used for sheet metals. Define various sheet metal operations like - shearing, cutting off, parting, blanking,. (3 marks)

► Define or explain 1 Understanding punching, notching, slitting, lancing, bending, drawing and squeezing.. (3 marks)

- **Define or explain Define welding and weldability. 1 Remembering Explain the features of welding and general. (3 marks)**

Module 3

- **Define or explain their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing, centering, drilling, plain turning, boring,. (3 marks)**

- **Define or explain reaming, chamfering taper turning and thread cutting. Identify the components of a of shaper and demonstrate the crank and slotted lever type 1 Understanding M3.03 quick return motion mechanism.. (3 marks)**

- **Define or explain Identify the components of a drilling machine.. (3 marks)**

- **Define or explain Define drilling, reaming and boring. 1 Understanding Identify the components of a Column and. (3 marks)**

Module 4

- **Define or explain general classification of metal forming 1 Understanding operations.. (3 marks)**

- **Define or explain Define forging. 1 Understanding Define forging, smith forging and machine. (3 marks)**

- **Define or explain 1 Understanding forging Demonstrate the machine forging of crank. (3 marks)**

- **Define or explain 1 Understanding shaft and connecting rod. Illustrate rolling and rolling mill. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Recall casting and pattern. 1 Remembering Outline the requirements of a pattern material.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **1 Understanding application of sheet metal in automobile. List metals used for sheet metal work and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **their functions, working and lathe 2 Understanding specifications. Define various lathe operations such as facing,**

Module 4

1-mark definition: State the meaning of **general classification of metal forming 1 Understanding operations..**

centering, drilling, plain turning, boring,.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION**Exam checklist****Before writing**

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.nptel.ac.in/courses/112105126/36
www.youtube.com/watch?v=T5gjkYvMg8A <http://www.youtube.com/watch?v=TeBX6cKKHWY>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTR syllabus PDF for Course 3052. Verify institutional updates against the current official curriculum.